

DBMS — LECTURE 2: FILE SYSTEM VS DBMS 📁⚡⚡

1 & 2. What is a File System & Why File Systems Fall Short?

- **File System:** Method used by OS to organize data in files, folders, drives & directories (e.g., **NTFS, FAT32, NFS, CIFS/SMB**).
- **Hierarchy:** Drive → Folder → Subfolder → File
- **Limitation:** Works great for 1 User managing local files. But modern multi-user apps (*Banking, IRCTC, Amazon*) need **centralized, simultaneous (concurrent) access**!

Client-Server Architecture:

[Server: Database + DBMS] ← (Thousands of Concurrent Users / Clients)

3, 4 & 5. Core Comparison: Data Retrieval & Location

📁 File System

- Must know exact file path (e.g., `C:\College\Students\student.txt`).
- To get 1 KB info, may need to load/scan entire 25 GB file! (High memory, slow retrieval).
- App directly depends on file organization.

🗄️ DBMS

- Uses **SQL Queries** (e.g. `SELECT * FROM Train WHERE ID=123;`).
- Retrieves **ONLY** required records accurately!
- Provides **Data Abstraction & Independence** (Logical vs Physical).

6 & 7. Metadata

- **Metadata = Data about Data**.
- **File System:** File Name, Path, Size, Created Date, Type.
- **DBMS:** Catalog / Data Dictionary storing info on **Tables, Columns, Data Types, Constraints, Users, Roles & Permissions**.

8, 9 & 10. Concurrency Control & The Single-Seat Problem ★

Concurrency: Multiple users/processes accessing shared data at the exact same time.

⚠️ THE RACE CONDITION PROBLEM:

If 1 seat remains and User A & User B try booking simultaneously:

Without control → **Lost updates, inconsistent data, double booking!**

DBMS Solution: Provides Concurrency Control via **Transactions, Locks, Isolation, & Serializability**!

11, 12 & 13. Security & Role-Based Access Control (RBAC)

- File Systems use OS file permissions (basic).
- DBMS provides **fine-grained, database-level authorization** (Roles & Privileges).

RBAC Example in University DB:

Student (Limited: Profile/Marks) | Faculty (More: Attendance/Marks Entry) | Admin/Dean (Full Access)

```
GRANT SELECT ON Student TO Faculty;
```

14-18. Data Redundancy & Data Integrity

- **Data Redundancy:** Unnecessary duplication of data. Causes: **Storage wastage, Inconsistency, Maintenance headaches**.
- **How DBMS Helps:** Database design, **Normalization**, Keys & Constraints (**PRIMARY KEY**, **FOREIGN KEY**, **NOT NULL**, **CHECK**).

🔥 **INTERVIEW TRICK: DOES PRIMARY KEY ELIMINATE ALL REDUNDANCY?**

NO! Primary key prevents duplicate key values in a table. Complete redundancy reduction requires **Good Database Design + Normalization + Constraints** !

19. Complete Comparison Table ★ (Exam Favorite!)

Feature	File System	DBMS
Data Storage	Files & Directories	Centralized Database
Data Access	File paths / File APIs	Query-based (SQL)
Concurrent Access	Limited / Basic mechanisms	Advanced Concurrency Control
Security	OS file permissions	Fine-grained DB Authorization & Roles
Redundancy & Integrity	High redundancy, low integrity	Reduced via Normalization & Constraints
Transaction & Recovery	Not supported natively	ACID Properties & Recovery Mechanisms

21, 22 & 23. Top 10 Advantages of DBMS Over File System ★

1. **Efficient Retrieval** (Query optimization)
2. **Data Abstraction** (Hides physical details)
3. **Concurrency Control** (Safe simultaneous access)
4. **Enhanced Security** (RBAC & fine-grained permissions)
5. **Reduced Redundancy** (Normalization & keys)
6. **Data Integrity** (Strict constraints like CHECK/FOREIGN KEY)
7. **Transaction Management** (ACID compliance)
8. **Backup & Recovery** (Automated failure recovery)
9. **Data Independence** (Separates logic from storage)
10. **Relationship Management** (Native table links)

25. Interview Quick Fire Q&A 🎯

1. **Why DBMS over File System?** File systems lack database-level concurrency control, query flexibility, ACID transactions, and fine-grained security required by multi-user apps.
2. **What is Concurrency Control?** DBMS mechanisms coordinating simultaneous transactions to prevent data inconsistency.
3. **What is Metadata?** Data about data (e.g. schema, constraints, data dictionary).
4. **Does File System have security?** Yes, OS-level permissions exist, but DBMS offers granular database-specific permissions.
5. **Does DBMS completely eliminate redundancy?** No, but normalization & good schema design minimize unnecessary duplication.

★ **THE 5 MOST IMPORTANT TAKEAWAYS:**

1. **File System = Files & Folders | DBMS = Shared Database Management.**

2. DBMS enables query-based, highly efficient data retrieval.
3. DBMS controls simultaneous access via Concurrency Control & Locks.
4. DBMS provides fine-grained Role-Based Access Control (RBAC).
5. DBMS maintains integrity & reduces redundancy via Normalization & Constraints.



END OF LECTURE 2 REVISION NOTES! 100